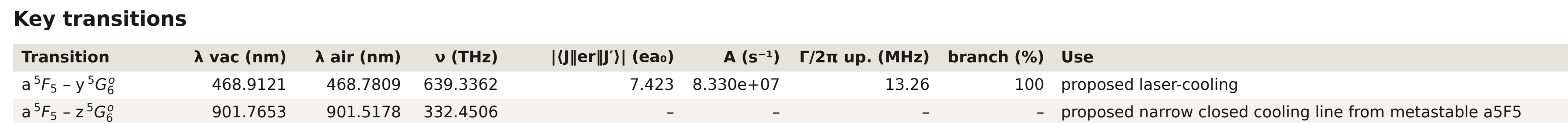


Zr⁺ ionisation limit 53507.83 cm⁻¹ = 6.63413 eV (186.889 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle f || e || i \rangle / \hbar$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.